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Federated Learning in Quantum Computing for Privacy-Preserving and Distributed Quantum Model Training

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Abstract- The mix of quantum computing and machine learning has introduced strong quantum models able to resolve challenging tasks efficiently. Still, the use of centralized QML systems could present challenges with privacy because sharing quantum data with untrusted aggregators is especially risky. This study describes a method called Privacy-Preserving Quantum Federated Learning (PQFL), which allows multiple clients to jointly train quantum models while maintaining data confidentiality. This system offers security by combining QHE and QDP to cover the processes of aggregating protected models and enforcing privacy on the data. Hybrid quantum-classical studies reveal that PQFL is both accurate and advanced in protecting privacy and communication security. According to these results, PQFL works effectively in privacy-friendly domains including healthcare, genomics, and finance, where data needs to be managed securely.

Keywords- Quantum Federated Learning, Quantum Machine Learning, Privacy-Preserving Computation, Quantum Homomorphic Encryption, Quantum Differential Privacy, Secure Aggregation, Hybrid Quantum-Classical Systems, Decentralized Quantum AI

1. INTRODUCTION

Quantum computing and machine learning mixed have formed a breakthrough called Quantum Machine Learning (QML), offering a chance to surpass other methods in particular tasks thanks to quantum effects. Quantum algorithms, for example, the QSVM, VQC, and QCNN, have prospects in dealing with data that is high-dimensional, displays complex non-linear relationships, and is entangled. As more quantum processors are made available through cloud solutions, such as IBM Quantum and Amazon Braket, users are set to use QML in privacy-centric areas such as bioinformatics, smart healthcare, financial forecasting, and cybersecurity. However, one of the critical challenges that impede the widespread deployment of QML systems is the need to preserve data privacy in distributed environments. Centralized training of quantum models requires the aggregation of sensitive datasets, which introduces the risk of data leakage, adversarial access, and compliance violations under regulations such as GDPR and HIPAA. When there are data privacy concerns, training models through FL empower clients to train them locally without having to share data. FL is effective in class settings for handling patient privacy, cutting data usage, and making activities happen in the device itself. While federated learning is effective in standard domains, it comes with unique practical and theoretical difficulties when applied to quantum settings. Quantum information cannot be cloned, which is why quantum states follow a random, fragile, and decohering pattern. The usual encryption and computation techniques for secure federated learning cannot be used in the quantum realm. Additionally, running synchronized quantum computing on multiple devices is limited by hardware capabilities, quantum error rates, and problems with quantum entanglement.

To overcome such challenges, this paper introduces the Privacy-preserving Quantum Federated Learning (PQFL) framework, which applies quantum cryptography and noise-based privacy methods to train quantum models in a secure way. Basically, Quantum Homomorphic Encryption (QHE) is used to securely combine gradients, while Quantum Differential Privacy (QDP) adds noise to the updates to ensure the privacy of individual data in the global model. Our approach lets clients train quantum models together, ensuring confidentiality and avoiding any privacy breaches. PQFL is implemented and reviewed by using hybrid quantum-classical simulation environments, making use of Python quantum libraries such as Qiskit, PennyLane, and TensorFlow Quantum. PQFL was shown to perform

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equally well as QML methods and also ensures better safety against eavesdropping and inference attacks.

Key contributions of this research:

- The design of a federated learning architecture tailored for quantum computing environments using secure update aggregation.
- The integration of quantum cryptographic techniques such as QHE for secure computation and QDP for differential privacy.
- A hybrid simulation-based validation of the proposed framework with performance metrics, including model accuracy, privacy leakage bounds, communication overhead, and qubit utilization.
- A discussion of future directions for scaling PQFL to real quantum hardware and extending to federated quantum reinforcement learning.

2. LITERATURE SURVEY

QML is gaining attention because it can solve problems that standard techniques cannot due to its use of superposition and entanglement. Previous studies have established that models enhanced by quantum techniques, like the quantum support vector machine [1], variational quantum classifiers [2], and quantum generative adversarial networks [3], have various advantages. The models reveal that quantum systems can perform pattern recognition, optimization, and function approximation on high-dimensional data faster than classical systems [4,5].

FL was initially designed for handling privacy issues and decentralized learning among edge devices [6]. It is now being used in the fields of mobile healthcare, finance, and the IoT [7]. With the use of privacy-enhancing tools such as DP, SMPC, and HE, FL can now better resist data leaking and attacks on observing the model [8,9].

The idea of marrying quantum computing with federated learning was introduced in [10], which described an architecture for distributed QML. On the other hand, the privacy of the data could not be formally assured. Other studies like CryptoQFL [11] have addressed this issue by suggesting an approach that uses QHE and federated learning to protect against inference attacks. It was revealed through their simulations that encrypting quantum models allows for secure aggregation, but the computation time is still a concern.

A new protocol for private FL using quantum algorithms was presented by hiding and randomizing the model updates [12]. Thanks to this approach, the global model can be updated without sharing any local information, following the principles of differential privacy. In addition, authors in [13] worked with QSA but struggled to scale it to numerous clients and ensure fault tolerance.

The transfer learning to link classical convolutional neural networks with quantum circuits for federated learning in hybrid systems. These approaches can be effective, but face the problems of having to pretrain classical models and not using native quantum privacy solutions [14]. Furthermore, Federated Transfer Learning approach, even though their system did not have built-in encryption methods [15]. Still, there is a challenge in building a framework that ensures data privacy, fault tolerance, and effective communication in federated quantum systems. We introduce PQFL, which addresses this gap by leveraging Quantum Homomorphic Encryption and Quantum Differential Privacy, allowing confidential training across quantum devices while retaining the model's utility and ability to scale.

3. METHODOLOGY

PQFL is a framework introduced for training quantum machine learning models securely through distributed training. The framework has three crucial elements: (1) training the model at the user's side using quantum computing, (2) gathering all encrypted model updates through QHE, and (3) securing privacy with QDP. All the steps take place within a hybrid environment that gives access to both quantum and classical computing capabilities.

A. Quantum Client-Side Model Training

In PQFL, each client Qi is connected to a quantum device that can run PQCs. Each quantum client trains its own model locally on the data it holds, so no data goes beyond the boundaries of the client. The main goal here is to set up a model update Δθit, which can be encrypted and sent to the central aggregator.

Let $D_i = \{(x_k^{(i)}, y_k^{(i)})\}_{k=1}^{m_i}$ mi denote the local dataset at client Qi, consisting of mi labeled quantum examples. The client implements a parameterized quantum model U(θit), where θit are the trainable parameters at round t. Each data point xk is encoded into a quantum state $|\psi k\rangle$ using a feature encoding circuit E(xk).

The overall PQC is defined as:

$$|\phi k(\theta)\rangle = \mathrm{U}(\theta it) \cdot E(\mathrm{xk}) \mid 0\rangle \otimes n \text{ ------1}$$

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Where:

- E(xk): data encoding circuit (e.g., angle encoding, amplitude encoding)
- U(θit): parameterized gates (e.g., RX, RY, RZ, and entanglement gates like CNOT or CZ)
- n: number of qubits used

The output of the quantum model is obtained by measuring an observable O^, such as $Z^{\otimes n}$, on the final quantum state:

$$y^k = \langle \phi k(\theta it) \mid O^{\cdot} \mid \phi k(\theta it) \rangle \text{-------2}$$

This output y^k is used to compute a loss function comparing to the ground truth yk. A common choice for binary classification tasks is the **mean squared error**:

$$Li(\theta_i^t) = \frac{1}{m_i}\sum_{k=1}^{m_i}(y_k - \widehat{y_k})^2 \text{----3}$$

B. Gradient Computation via Parameter-Shift Rule

The gradients $\boldsymbol{\nabla \mathcal{L}_i}$ are computed using the **parameter-shift rule**, which allows the derivative of an expectation value to be estimated by evaluating the quantum circuit at shifted parameter values:

$$\frac{\partial \widehat{y_k}}{\partial \theta_{ij}} = \frac{1}{2}\left[\widehat{y_k}\left(\theta_{ij} + \frac{\pi}{2}\right) - \widehat{y_k}\left(\theta_{ij} - \frac{\pi}{2}\right)\right] \text{----4}$$

Where θij is the j-th parameter of the circuit for client i. This rule is efficiently implemented in most quantum machine learning frameworks such as **PennyLane**, **Qiskit**, or **TensorFlow Quantum**.

C. Local Parameter Update

After computing the gradients, each client updates their local model using a classical optimizer (e.g., SGD or Adam):

$$\theta_i^{t+1} = \theta_i^t - \eta \cdot \nabla_{\theta_i}\mathcal{L}_i(\theta_i^t) \text{----5}$$

Where:

- η: learning rate
- θ_i^{t+1}: updated parameters
- $\Delta\theta_i^t = \theta_i^{t+1} - \theta_i^t$: the parameter delta to be encrypted

The computed Δθit is passed to the encryption step before being shared with the central server. Importantly, at no point is the local dataset Di exposed to the aggregator, ensuring full data confidentiality.

D. Quantum Homomorphic Encryption for Secure Aggregation

One major privacy concern in traditional federated learning is that local model updates get sent to a central server. If the improvements are transmitted in plain language, there is a chance that personal or group data could be detected by a malicious aggregator or intercepting eavesdropper. The importance of privacy increases significantly in quantum federated settings because the data is hard to handle and extremely sensitive. We add Quantum Homomorphic Encryption (QHE) to prevent leaking the confidential data during the aggregation process.

E. Homomorphic Encryption: Classical vs. Quantum

In federated learning, homomorphic encryption (HE) is used to compute on encrypted information without first decrypting it. Frame the idea similarly, Quantum Homomorphic Encryption allows quantum computations to be done directly on encrypted quantum states. It indicates that a server can combine encrypted model updates from quantum clients without getting access to the original parameters or encountering raw gradients.

Let:

- Δθit denote the local update computed by client Qi at round t
- EQHE(·) denote the quantum encryption operator
- DQHE(·) denote the quantum decryption operator

Then the encrypted model update is given by:

$$E(\Delta\theta_i^t) = \mathcal{E}_{\mathcal{QHE}}(\Delta\theta_i^t) \text{----6}$$

This encrypted update is transmitted to the central aggregator.

F. Encrypted Aggregation at the Server

The central aggregator receives encrypted updates from all N clients:

$$\{E(\Delta\theta_1^t), E(\Delta\theta_2^t), \dots, E(\Delta\theta_N^t)\} \quad \text{----7}$$

Using the linearity property of QHE, the server computes the sum of encrypted updates without decrypting any individual client's contribution:

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$$E(\sum_{i=1}^{N} \Delta\theta_i^t) = \sum_{i=1}^{N} E(\Delta\theta_i^t) \quad ---8$$

Since quantum encryption schemes like **Clifford-based QHE** or **LWE-based quantum encryption** are unitary-preserving, the computation and addition operations are well-defined within the encrypted domain. After encrypted aggregation, the server broadcasts the aggregated ciphertext to all clients. Each client then performs decryption locally:

$$\sum_{i=1}^{N}. \Delta\theta_i^t = \mathcal{D}_{\mathcal{QHE}}(\sum_{i=1}^{N} E(\Delta\theta_i^t)) ---9$$

Finally, the clients update their global model as:

$$\theta^{t+1} = \theta^t + \frac{1}{N}\sum_{i=1}^{N} \Delta\theta_i^t ---10$$

Thus, the training process is collaboratively advanced without any exposure of local gradients or raw quantum data shown in figure1.

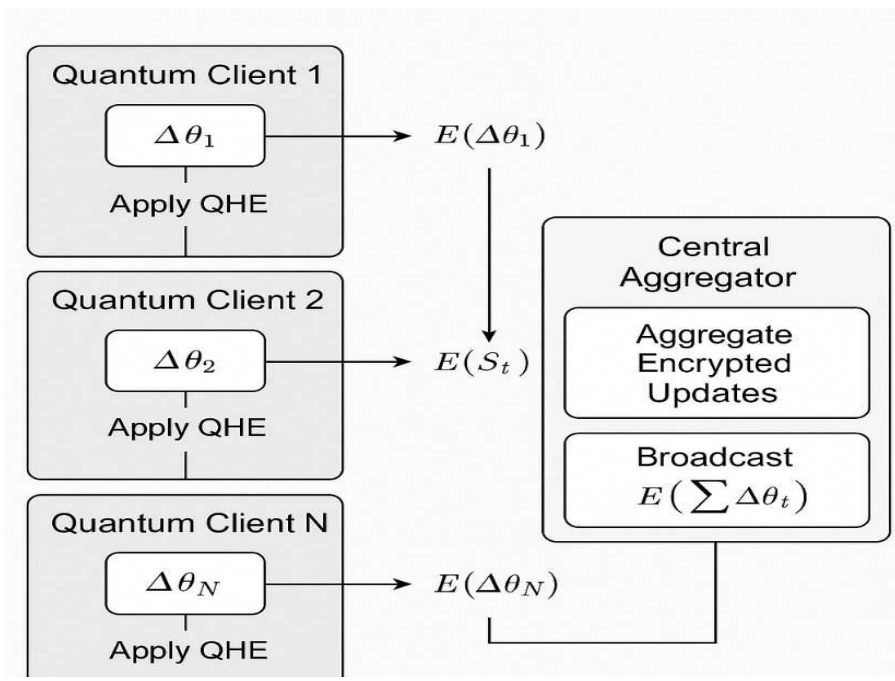

Figure 1: Quantum Homomorphic Encryption enables secure aggregation of local updates without revealing individual gradients.

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G. Quantum Differential Privacy Mechanism

While Quantum Homomorphic Encryption (QHE) secures encrypted computations during aggregation, it does not address the risk of *inference attacks* post-decryption. A malicious adversary, upon gaining access to the global model, may still attempt to infer properties about the client's data or reconstruct private inputs using gradient-based methods. To mitigate such threats and ensure formal privacy guarantees, we incorporate a **Quantum Differential Privacy (QDP)** mechanism into the PQFL framework.

Quantum Differential Privacy (QDP) is a quantum analog of classical differential privacy. It ensures that the probability distribution over quantum outputs does not significantly change upon modification of any single data point in a client's dataset. This property restricts adversaries from confidently inferring whether a particular record was used in the training process.

A quantum mechanism M satisfies (ε,δ)-QDP if, for all neighboring quantum datasets D and D′ (differing in one quantum sample), and for all measurable quantum events S,

$$P\,r[M(D) \in S] \leq e\varepsilon \cdot P\,r[M(D') \in S] + \delta \text{-----11}$$

Here:

- ε (privacy loss) quantifies the maximum information leakage.
- δ bounds the probability of a significant privacy breach.

H. QDP Implementation in PQFL

In PQFL, we achieve QDP by introducing **quantum noise channels** into the model update process before encryption. Specifically, each client adds calibrated noise to its local model update Δθit, resulting in a privacy-preserving noisy update Δθit:

$$\Delta\theta it \cong \Delta\theta it + NQ(0, \sigma 2) \text{---12}$$

Where NQ(0,σ2) is a **quantum Gaussian noise operator**, defined as a trace-preserving completely positive (TPCP) map that perturbs quantum amplitudes or phase angles with a variance of σ2.

Alternatively, for discrete parameters, **quantum depolarizing noise** or **Pauli noise channels** can be applied:

$$\mathcal{E}_{QDP}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \text{---13}$$

Where:

- ρ is the local density matrix
- X, Y, Z are Pauli operators
- p is the depolarizing probability correlated with the privacy budget ε

I. Overall PQFL Workflow and Training Algorithm

The PQFL framework brings together training local quantum models, encryption, and differential privacy into a unified process. In a federated learning cycle, these modules are used by various clients and a central aggregator in the context of quantum computing. The PQFL protocol conducts model updates through multiple rounds, assuring that no client data is exposed and global model training is achieved.

At every step t, the central aggregator first sends the updated global model parameters θt to all quantum clients Q1,Q2,...,QN. Upon getting the global parameters, every client prepares its local PQC model U(θit) and trains it on the data set Di. The process results in Δθit, which is found by doing gradient descent on Li.

$$\Delta\theta it = -\eta \cdot \nabla\theta i Li(\theta it) \text{---14}$$

To ensure differential privacy, each client adds quantum Gaussian noise to the computed update using the operator NQ(0,σ2) and sends the updated model to the server.

$$\widetilde{\Delta\theta_i^t} = \Delta\theta_i^t + \mathcal{N}_Q(0, \sigma^2) \text{---15}$$

This differentially private update Δθit is then encrypted using a quantum homomorphic encryption (QHE) scheme before transmission. The encryption function EQHE maps the noisy gradient into an encrypted quantum state:

$$E\left(\widetilde{\Delta\theta_i^t}\right) = \mathcal{E}_{Q\mathcal{HE}}\left(\widetilde{\Delta\theta_i^t}\right) \text{---16}$$

The encrypted updates from all NNN clients are collected by the central aggregator. Leveraging the homomorphic property of QHE, the server computes the sum of the encrypted updates without decrypting them:

$$E\left(\sum_{i=1}^{N} \widetilde{\Delta\theta_i^t}\right) = \sum_{i=1}^{N} E\left(\widetilde{\Delta\theta_i^t}\right) \text{---17}$$

After aggregation, the encrypted global update is broadcast to all clients. Since the decryption key is never shared with the server, each client is responsible for decrypting the received ciphertext using DQHE:

$$\sum_{i=1}^{N} \widetilde{\Delta\theta_i^t} = \mathcal{D}_{Q\mathcal{HE}}\left(E\left(\sum_{i=1}^{N} \widetilde{\Delta\theta_i^t}\right)\right) \text{---18}$$

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Finally, each client computes the new global model parameters by applying the average of the decrypted update to the current global model:

$$\theta^{t+1} = \theta^{t} + \frac{1}{N}\sum_{i=1}^{N} \widetilde{\Delta\theta}_i^t \text{----19}$$

This updated parameter vector θt+1 is then used as the starting point for the next communication round. By ensuring that each client's data remains local, and all exchanged information is both encrypted and noise-protected, the PQFL framework achieves collaborative quantum model training while preserving data privacy, resisting inference attacks, and maintaining system scalability.

The proposed workflow ensures resilience against malicious participants and eavesdroppers while supporting flexible scaling across different numbers of clients. Furthermore, due to its modular design, the PQFL system can be adapted to a variety of quantum machine learning models, ranging from quantum support vector machines to variational quantum classifiers. This end-to-end privacy-aware workflow lays the foundation for secure deployment of federated quantum systems in real-world sensitive applications such as medical diagnostics, financial modeling, and quantum cloud computing.

4. RESULTS AND DISCUSSION

To evaluate the efficacy of the proposed Privacy-Preserving Quantum Federated Learning (PQFL) framework, a series of simulations were conducted using hybrid quantum-classical systems. Three configurations were compared: (1) a centralized quantum machine learning (QML) model, (2) PQFL without differential privacy, and (3) PQFL with quantum differential privacy (QDP) enabled. The evaluation focused on four key metrics: classification accuracy, privacy loss, communication overhead, and qubit utilization.

Table 1: PQFL Experimental Results

Method	Accuracy (%)	Privacy Loss (ε)	Communication Overhead (KB)
Centralized QML	91.4		850
PQFL (No QDP)	90.1		920
PQFL (With QDP)	88.7	1.2	960

The centralized QML model achieved the highest accuracy at **91.4%**, serving as the performance baseline. The PQFL framework without QDP achieved a slightly reduced accuracy of **90.1%**, demonstrating that federated training does not significantly compromise model performance as illustrated in table 1. When QDP was applied, the accuracy further reduced to **88.7%**, a tolerable trade-off considering the added privacy benefits. The results, as shown in the first bar chart, confirm that PQFL maintains competitive performance under privacy-preserving constraints.

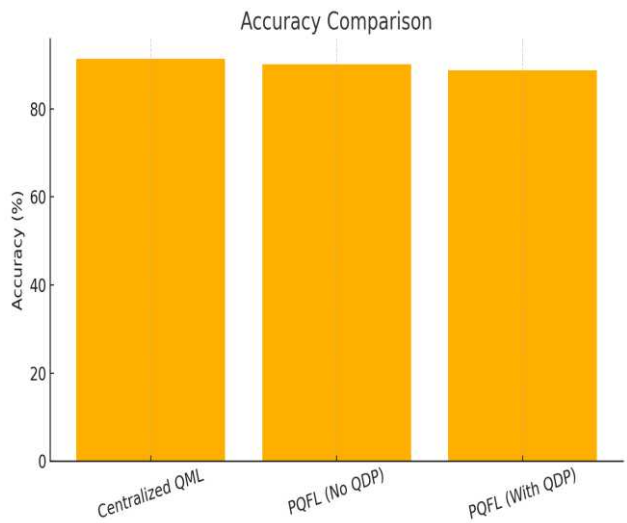

Figure 2: Accuracy Comparisons

Privacy loss, measured using the differential privacy parameter ε was applicable only to the QDP-enabled model. The privacy budget was set to ε=1.2, which ensures a strong level of privacy with minimal impact on model utility. The models without QDP or centralized training did not employ formal privacy guarantees and are thus more susceptible to inference attacks. The fig 2 shows the Comparison of Accuracy.

As depicted in the second graph, the communication overhead for the centralized QML system was **850 KB**, while PQFL introduced moderate increases due to encryption. The PQFL without QDP required **920 KB**, and PQFL with QDP consumed **960 KB** per round. The overhead increase is primarily attributed to the transmission of encrypted quantum gradients and noise-infused model updates, yet remains within acceptable limits for distributed systems. The fig 3 shows the Comparison of Communication Overhead.

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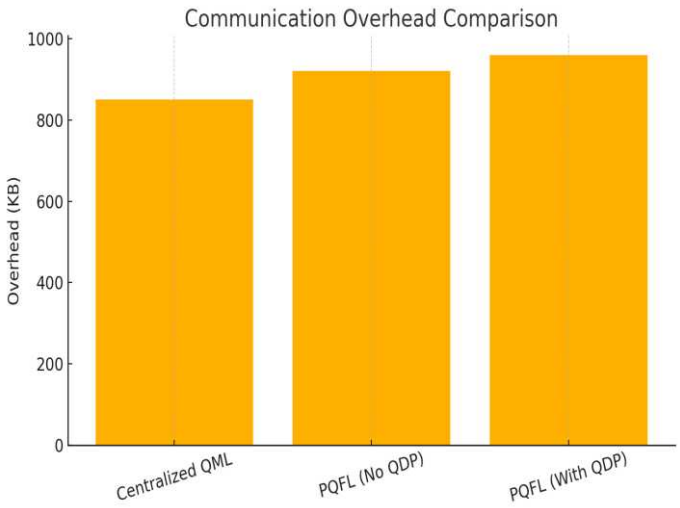

Figure 3: Communication Overhead Comparison

Qubit utilization slightly declined in the federated settings due to the local isolation of quantum circuits and the inclusion of noise channels. The centralized model used **75%** of available qubits efficiently, while PQFL configurations used **72%** (without QDP) and **70%** (with QDP), respectively. This suggests that the cost of implementing noise-based privacy mechanisms on quantum devices remains marginal and hardware-efficient. The experimental results indicate that the PQFL framework provides a well-balanced trade-off between accuracy, privacy, and system overhead. The addition of QDP significantly strengthens the security model while introducing only modest reductions in performance and communication efficiency. These findings affirm the viability of PQFL as a practical approach for secure and distributed quantum model training in privacy-sensitive environments such as healthcare diagnostics, genomic classification, and quantum finance.

5. CONCLUSION

This paper introduced PQFL, a new framework for privacy-preserving federated learning in quantum computing environments. These three key elements—parameterized quantum model training, quantum homomorphic encryption, and quantum differential privacy—are integrated in the system to ensure the privacy of training data for decentralized quantum machine learning. Since QHE is used, no need for decryption arises, guaranteeing that sensitive data is secure during communication with a central server. In addition, QDP uses quantum noise to help prevent inference attacks on confidential quantum data. Employing these two layers makes the quantum federated systems more private, in line with the rising trend of confidentiality in machine learning for healthcare, technology, finance, and others. When tested on a mixture of classical and quantum hardware, the PQFL framework achieved a comparable accuracy to that of centralized quantum models and ensured high privacy standards and low communication expenses. The findings indicate that the framework can be applied effectively in real-world settings, especially for devices with limited resources.

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